

Lecture 13: Linear Coupling and Acceleration

TTIC 31070 / CMSC 35470 / BUSF 36903 / STAT 31015
Convex Optimization

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From the Lower Bound to the Algorithm

Lecture 12 (lower bound). For smooth L -smooth convex optimization in radius R :

$$\inf_A \sup_f [f(x_T) - f^*] \geq \Omega\left(\frac{LR^2}{T^2}\right).$$

Where we are. Gradient descent gives $O(LR^2/T)$ — off by a factor of T .

Today. Construct an algorithm achieving $O(LR^2/T^2)$ — matching the lower bound.

Viewpoint. *Linear coupling* (Allen-Zhu & Orecchia, ITCS 2017): one gradient step gives primal progress, one mirror step gives dual progress, and a *coupled query point* makes the two terms match.

Color convention used today. y sequence (gradient step), z sequence (mirror step), x sequence (queries).

Euclidean Preview: Two Uses of the Same Gradient

Setup (intuition only, Euclidean). Query at x , set $g := \nabla f(x)$.

Use 1: Gradient step.

$$y^+ := x - \frac{1}{L}g.$$

Smoothness gives

$$f(y^+) \leq f(x) - \frac{1}{2L}\|g\|_2^2.$$

Large gradient $\Rightarrow$ instant objective decrease.

Use 2: Mirror step.

$$z^+ := z - \alpha g.$$

For any comparator u ,

$$\begin{aligned} \alpha \langle g, z - u \rangle &\leq \frac{1}{2}\|u - z\|_2^2 - \frac{1}{2}\|u - z^+\|_2^2 \\ &\quad + \frac{\alpha^2}{2}\|g\|_2^2. \end{aligned}$$

Small gradient $\Rightarrow$ linear signal still becomes telescoping distance decrease.

Two regimes, one gradient. Gradient step likes $\|g\|$ *big*; mirror step is fine when $\|g\|$ is *small*. **Acceleration couples them so no case split is needed.**

Primal Progress

Lemma 13.1 (Primal progress). f is L -smooth on X ; $\mathbf{x} \in X$, $\mathbf{g} = \nabla f(\mathbf{x})$, and

$$\mathbf{y}^+ \in \operatorname{argmin}_{\mathbf{q} \in X} \left\{ \langle \mathbf{g}, \mathbf{q} - \mathbf{x} \rangle + \frac{L}{2} \|\mathbf{q} - \mathbf{x}\|^2 \right\}.$$

Then for **every** $\mathbf{v} \in X$:

$$f(\mathbf{y}^+) \leq f(\mathbf{x}) + \langle \mathbf{g}, \mathbf{v} - \mathbf{x} \rangle + \frac{L}{2} \|\mathbf{v} - \mathbf{x}\|^2.$$

We will eventually pick $\mathbf{v}$ to be the auxiliary point that makes the mirror displacement match a gradient-model displacement.

Dual Progress

Lemma 13.2 (Dual progress). X convex; h convex and differentiable on a neighborhood of X ; $z \in X$, $g \in E^*$, $\alpha > 0$, and

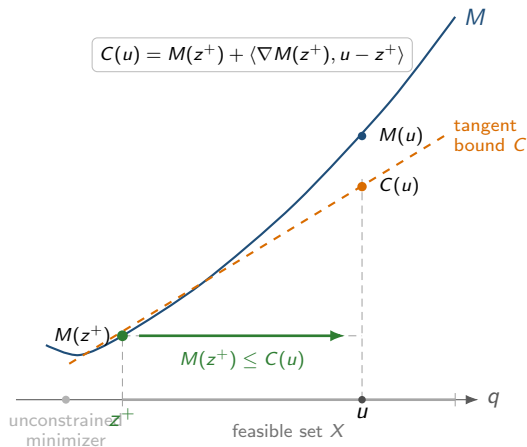
$$z^+ \in \operatorname{argmin}_{q \in X} \left\{ \alpha \langle g, q - z \rangle + D_h(q, z) \right\}.$$

Then for **every** $u \in X$:

$$\alpha \langle g, z^+ - z \rangle + D_h(z^+, z) \leq \alpha \langle g, u - z \rangle + D_h(u, z) - D_h(u, z^+).$$

The pair $D_h(u, z) - D_h(u, z^+)$ is the **telescoping potential drop**.

Three-Point Inequality: Tangent-Bound View



Recall the three-point inequality for

$$M(q) := \alpha \langle g, q - z \rangle + D_h(q, z).$$

The mirror step chooses

$$z^+ \in \operatorname{argmin}_{q \in X} M(q).$$

At a constrained minimizer, $\nabla M(z^+)$ need not vanish. The first-order condition gives

$$\langle \nabla M(z^+), u - z^+ \rangle \geq 0, \quad u \in X.$$

Thus

$$M(z^+) \leq C(u)$$

$$:= M(z^+) + \langle \nabla M(z^+), u - z^+ \rangle.$$

Expanding this tangent value gives

$$C(u) = \alpha \langle g, u - z \rangle + D_h(u, z)$$

$$- D_h(u, z^+),$$

which is the three-point inequality used in Lemma 13.2.

Linear Coupling: The Update

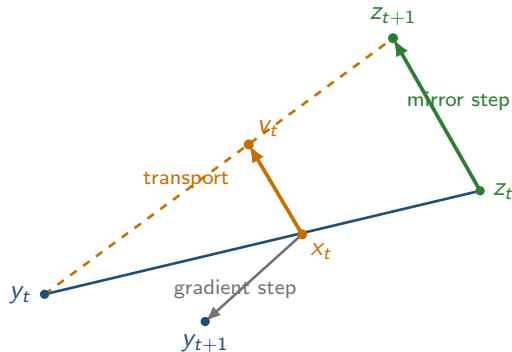
Definition 13.1 (Linear coupling step). Given $y_t, z_t \in X$ and $\tau_t \in (0, 1]$:

- ▶ **Coupled query.** $x_t := (1 - \tau_t)y_t + \tau_t z_t$, $g_t := \nabla f(x_t)$.
- ▶ **Gradient step (at x_t).** $y_{t+1} = \operatorname{argmin}_{q \in X} \left\{ \langle g_t, q - x_t \rangle + \frac{L}{2} \|q - x_t\|^2 \right\}$.
- ▶ **Mirror step (at z_t).** $z_{t+1} = \operatorname{argmin}_{q \in X} \left\{ \frac{1}{\tau_t} \langle g_t, q - z_t \rangle + \frac{L}{\rho} D_h(q, z_t) \right\}$.

Three sequences, one gradient per step.

- ▶ x_t : where we evaluate ∇f — a *convex combination* of y_t and z_t .
- ▶ y_t : tracks gradient-step descent; the *output sequence*.
- ▶ z_t : tracks mirror-step distance; the *dual-progress sequence*.

Linear Coupling: Geometry and the Auxiliary v_t



Auxiliary point. $v_t := (1 - \tau_t)y_t + \tau_t z_{t+1}$ (replace $z_t \rightarrow z_{t+1}$, same τ_t).

Transport identity. $v_t - x_t = \tau_t(z_{t+1} - z_t)$, $z_t - z_{t+1} = \frac{x_t - v_t}{\tau_t}$.

Mirror displacement at z_t = gradient-model displacement at x_t (scaled by τ_t). **This couples primal and dual progress.**

One-Step Coupling Inequality

Theorem 13.3 (One-step linear-coupling). f, h differentiable; f convex and L -smooth; h ρ -strongly convex. With y, z, τ, x, y^+, z^+ as in Definition 13.1, for every $u \in X$:

$$\frac{\rho}{L\tau^2}(f(y^+) - f(u)) + D_h(u, z^+) \leq \frac{\rho(1-\tau)}{L\tau^2}(f(y) - f(u)) + D_h(u, z).$$

Let $\Phi(y, z; \tau) := \frac{\rho}{L\tau^2}(f(y) - f(u)) + D_h(u, z)$. The inequality says the new state's potential at τ is dominated by the old state's potential at τ , but with the function-gap weight *shrunk* by $(1 - \tau)$.

Why this is acceleration.

- ▶ GD-style descent gives f -decrease at y^+ weighted by the *unshrunk* potential.
- ▶ Coupling produces an extra $(1 - \tau)$ factor on the *old* f -gap — exactly the slack a telescoping sum needs.

Proof Piece A: Primal Progress

Set the auxiliary comparator

$$v := (1 - \tau)y + \tau z^+$$

and define the gradient-model residual

$$R_g := \langle g, v - x \rangle + \frac{L}{2} \|v - x\|^2.$$

Lemma 13.1, applied with this comparator v , gives

$$\boxed{f(y^+) \leq f(x) + R_g.} \tag{A}$$

Role of A. This is the only place where smoothness enters. The gradient step at x competes with the quadratic upper model evaluated at the transported point v .

Proof Piece B: Dual Progress

Apply Lemma 13.2 to the mirror step in Definition 13.1, i.e. to the mirror function $\tilde{h} = (L/\rho)h$ and scalar $\alpha = 1/\tau$. Since $D_{\tilde{h}} = (L/\rho)D_h$,

$$\frac{1}{\tau} \langle g, z^+ - z \rangle + \frac{L}{\rho} D_h(z^+, z) \leq \frac{1}{\tau} \langle g, u - z \rangle + \frac{L}{\rho} (D_h(u, z) - D_h(u, z^+)).$$

Multiplying by τ^2 gives a dual-progress bound at the scale needed for the coupling proof. Since h is ρ -strongly convex,

$$\frac{L\tau^2}{\rho} D_h(z^+, z) \geq \frac{L\tau^2}{2} \|z^+ - z\|^2.$$

Therefore, with $R_z := \tau \langle g, z^+ - z \rangle + \frac{L\tau^2}{2} \|z^+ - z\|^2$,

$$R_z \leq \tau \langle g, u - z \rangle + \frac{L\tau^2}{\rho} (D_h(u, z) - D_h(u, z^+)).$$

 (B)

Proof Piece C: Coupling via Convexity

Convexity at the query point x gives

$$f(x) - f(u) \leq \langle g, x - u \rangle = \langle g, x - z \rangle + \langle g, z - u \rangle.$$

Since $x = (1 - \tau)y + \tau z$,

$$x - z = \frac{1 - \tau}{\tau}(y - x).$$

Convexity again gives $\langle g, y - x \rangle \leq f(y) - f(x)$. Therefore

$$\tau(f(x) - f(u)) \leq (1 - \tau)(f(y) - f(x)) + \tau \langle g, z - u \rangle. \quad (C)$$

Role of C. It turns the comparator gap at x into an old y -gap plus the linear term that will cancel dual progress.

Proof Piece D: Transport Identity

The auxiliary point uses the same convex-combination weight as the query point:

$$\mathbf{x} = (1 - \tau)\mathbf{y} + \tau\mathbf{z}, \quad \mathbf{v} = (1 - \tau)\mathbf{y} + \tau\mathbf{z}^+.$$

Subtracting gives

$$\mathbf{v} - \mathbf{x} = \tau(\mathbf{z}^+ - \mathbf{z}).$$

Hence the two residuals from A and B are actually the same quantity:

$$\boxed{\langle \mathbf{g}, \mathbf{v} - \mathbf{x} \rangle + \frac{L}{2} \|\mathbf{v} - \mathbf{x}\|^2 = \tau \langle \mathbf{g}, \mathbf{z}^+ - \mathbf{z} \rangle + \frac{L\tau^2}{2} \|\mathbf{z}^+ - \mathbf{z}\|^2.} \quad (\text{D})$$

Role of D. The mirror displacement at $\mathbf{z}$ becomes the gradient-model displacement at $\mathbf{x}$. This is the actual “linear coupling” step in the proof.

Proof Recipe: Combine the Pieces

Add $(A) + (B) + (C)$, then use (D) . The two non-telescoping leftovers cancel:

$$\tau \langle g, u - z \rangle + \tau \langle g, z - u \rangle = 0, \quad R_g = R_z.$$

What remains is

$$f(y^+) - f(u) + \frac{L\tau^2}{\rho} D_h(u, z^+) \leq (1 - \tau)(f(y) - f(u)) + \frac{L\tau^2}{\rho} D_h(u, z).$$

Multiplying by $\rho/(L\tau^2)$ gives Theorem 13.3:

$$\frac{\rho}{L\tau^2} (f(y^+) - f(u)) + D_h(u, z^+) \leq \frac{\rho(1 - \tau)}{L\tau^2} (f(y) - f(u)) + D_h(u, z).$$

Telescoping: Choosing τ_t

Potential. Pick $u = x^*$. Define

$$\Phi_t := \frac{\rho}{L\tau_t^2} (f(y_{t+1}) - f^*) + D_h(x^*, z_{t+1}).$$

One-step (Theorem 13.3).

$$\Phi_t \leq \frac{\rho(1 - \tau_t)}{L\tau_t^2} (f(y_t) - f^*) + D_h(x^*, z_t).$$

Telescope condition. Need $\frac{1 - \tau_t}{\tau_t^2} \leq \frac{1}{\tau_{t-1}^2}$ so the right side is dominated by Φ_{t-1} .

Choice $\tau_t = \frac{2}{t+2}$. Then

$$\frac{1 - \tau_t}{\tau_t^2} = \frac{t(t+2)}{4} \leq \frac{(t+1)^2}{4} = \frac{1}{\tau_{t-1}^2}. \checkmark$$

For $t = 0$: $\tau_0 = 1 \Rightarrow$ coefficient is 0, so the chain starts cleanly.

Accelerated Smooth Convex Rate

Theorem 13.4 (Accelerated rate). f convex L -smooth; h ρ -strongly convex. With $\tau_t = 2/(t+2)$ and $y_0, z_0 \in X$:

$$f(y_T) - f(x^*) \leq \frac{4L D_h(x^*, z_0)}{\rho(T+1)^2}.$$

Euclidean specialization. $X = \mathbb{R}^n$, $h(z) = \frac{1}{2}\|z\|_2^2$, $\rho = 1$:

$$f(y_T) - f(x^*) \leq \frac{2L\|x^* - z_0\|_2^2}{(T+1)^2}.$$

This matches Lecture 12's lower bound $\Omega(LR^2/T^2)$ up to constants.

✓

Proof sketch. Telescope $\Phi_t \leq \Phi_{t-1}$ from $t = 0$ to $T-1$, then divide out $\rho/(L\tau_{T-1}^2) = \rho(T+1)^2/(4L)$ from the function-gap weight at T . Initial $\Phi_{-1} = D_h(x^*, z_0)$ since $\tau_0 = 1$ makes the convex-combination weight zero.

Smooth Strongly Convex: Restart

Idea. Run the convex accelerated method for N steps, restart with new $z_0 = y_N$.

Each epoch (use Theorem 13.4 + extra Bregman bound $D_h(a, b) \leq \frac{M}{2} \|a - b\|^2$):

$$f(w_{k+1}) - f^* \leq \frac{2LM}{\rho(N+1)^2} \|x^* - w_k\|^2.$$

Strong convexity of f converts distance to gap: $\|x^* - w_k\|^2 \leq \frac{2}{\mu} (f(w_k) - f^*)$. Hence

$$f(w_{k+1}) - f^* \leq \frac{4LM}{\rho\mu(N+1)^2} (f(w_k) - f^*).$$

Choose $N+1 \geq \sqrt{8LM/(\rho\mu)}$: contraction $\leq 1/2$ per epoch.

Smooth Strongly Convex: Accelerated Linear Rate

Theorem 13.5 (Restarted acceleration). f convex L -smooth and μ -strongly convex; h ρ -strongly convex with $D_h(a, b) \leq \frac{M}{2} \|a - b\|^2$. With $N + 1 \asymp \sqrt{LM/(\rho\mu)}$ steps per epoch and K epochs:

$$f(w_K) - f^* \leq 2^{-K} (f(w_0) - f^*).$$

Total complexity: $O(\sqrt{LM/(\rho\mu)} \log(1/\varepsilon))$ gradient evaluations.

In Euclidean ($\rho = M = 1$): $O(\sqrt{L/\mu} \log(1/\varepsilon))$.

Compare with non-accelerated.

- ▶ GD (smooth strongly convex): $O(\kappa \log(1/\varepsilon))$, $\kappa = L/\mu$.
- ▶ Restarted AGD: $O(\sqrt{\kappa} \log(1/\varepsilon))$. **Improvement:** $\sqrt{\kappa}$.

This matches Lecture 12's lower bound $\Omega(\sqrt{\kappa} \log(1/\varepsilon))$.



Why Acceleration Works (Recap)

In one sentence. Same gradient at x_t feeds two updates that produce *cancellable residuals*.

- ▶ **Gradient step at x_t** gives primal progress through a quadratic upper model at any v .
- ▶ **Mirror step at z_t** gives dual progress through a Bregman potential drop at any u .
- ▶ **Coupled query $x_t = (1 - \tau_t)y_t + \tau_t z_t$** + transport identity: the mirror residual at z_t equals the gradient-model residual at x_t when v is chosen as the auxiliary $v_t = (1 - \tau_t)y_t + \tau_t z_{t+1}$.

What changes vs. GD. Not the gradient information, not the smoothness, not the strong convexity. **Only how the same gradient is used twice:** primal progress + dual progress.

Cost: one gradient per step (same as GD), three sequences to maintain (vs. one).

Euclidean Specialization: Nesterov Momentum Form

Setting. $X = \mathbb{R}^n$, $h(z) = \frac{1}{2}\|z\|_2^2$, $\rho = 1$.

Linear coupling becomes: $y_{t+1} = x_t - \frac{1}{L}g_t$, $z_{t+1} = z_t - \frac{1}{L\tau_t}g_t$.

Algebra: $z_{t+1} = y_t + \frac{1}{\tau_t}(y_{t+1} - y_t)$, so

$$x_{t+1} = (1 - \tau_{t+1})y_{t+1} + \tau_{t+1}z_{t+1} = y_{t+1} + \beta_t(y_{t+1} - y_t),$$

with $\beta_t = \tau_{t+1}(1 - \tau_t)/\tau_t = (s_t - 1)/s_{t+1}$, $s_t = 1/\tau_t$.

Nesterov's Accelerated Gradient (1983).

$$\tilde{x}_{t+1} = \tilde{y}_{t+1} + \beta_t(\tilde{y}_{t+1} - \tilde{y}_t), \quad \tilde{y}_{t+2} = \tilde{x}_{t+1} - \frac{1}{L}\nabla f(\tilde{x}_{t+1}).$$

Same algorithm, different variables. Linear coupling $\Leftrightarrow$ NAG via $(x, y, z) \leftrightarrow (\tilde{x}, \tilde{y})$
with z implicit.

Summary

Linear coupling. Three sequences y_t , z_t , x_t , one gradient per step.

- ▶ Gradient step at x_t (descent).
- ▶ Mirror step at z_t (telescoping potential).
- ▶ Coupled query $x_t = (1 - \tau_t)y_t + \tau_t z_t$ + transport identity $\Rightarrow$ residuals cancel.

Rates.

- ▶ Smooth convex: $f(y_T) - f^* \leq 4LD_h(x^*, z_0)/(\rho(T+1)^2)$. $O(LR^2/T^2)$ — **tight**.
- ▶ Smooth strongly convex (restart): $O(\sqrt{\kappa} \log(1/\varepsilon))$. $\sqrt{\kappa}$ vs κ for **GD**.

Lecture 12 + Lecture 13 = tight. Acceleration is exactly the algorithm that closes the smooth Euclidean lower bound.

Next. Second-order methods: Hessians, local quadratic models, and Newton's method.

Bibliographic Notes

- ▶ **Nesterov (1983)**. Original accelerated gradient method via estimating sequences.
- ▶ **Nesterov (Introductory Lectures, 2004)**. Textbook treatment; constant-step and adaptive variants for smooth convex / smooth strongly convex.
- ▶ **Allen-Zhu & Orecchia (ITCS 2017)**. Linear coupling viewpoint: the coupled query + primal/dual progress derivation used today.
- ▶ **Bubeck (Convex Optimization: Algorithms and Complexity, 2015)**. Modern survey; Section 3.7 gives a compact linear-coupling treatment.